Lesson plan · for the teacher

Charts that tell the truth

Data science · Level 1 · Lesson 3

Time
40 min

Materials
Squared paper and coloured pencils; A small table of school data; A phone or tablet with Darsi, for the lab

Learning goals

- Choose bar, line, pie or scatter for a question.
- Spot a cut axis.
- Read a chart's title, axes and units.

Activity: Same data, four charts

1. Warm-up (5 min): Show the rainfall pie chart and ask what question it fails to answer.
2. Teach (8 min): Sketch a bar, a line, a pie and a scatter chart, each with the question it answers. Then show the cut-axis trick with the 70% and 74% bars.
3. Draw (15 min): Give every group the same table and a chart type. One group secretly draws a misleading version.
4. Gallery (7 min): Pin up the charts. Pupils vote on the clearest one and find the misleading chart, saying exactly what is wrong with it.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence, with a number in it. Then give the exit ticket.

Discussion

- Which chart would you use for attendance over a term?
- Why do bars have to start at zero, but lines don't always?
- Where have you seen a misleading chart?

Exit ticket

- Draw a tiny honest bar chart of two numbers, with a title and labelled axes.

Try it in class tomorrow
Give groups the same small table and ask each for a different chart. Pin them up and ask which chart answers the question best.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **A poster's bar chart has an axis starting at 90. What should you check first?**

◯ The title font ◯ The colours ◯ Where the axis starts

2 **Match each question to its chart.**

Which fruit does our class like best?	• • • •	Line chart
How did rainfall change this year?	• • • •	Bar chart
How is the school budget shared?	• • • •	Scatter chart
Do pupils who read more spell better?	• • • •	Pie chart

3 **This program draws a text bar chart of the pupils' favourite fruit. What is the second line it prints?**

```
fruits = ["Mango", "Banana", "Guava"]
votes = [5, 3, 2]
for i in range(3):
    print(fruits[i], "#" * votes[i])
```

◯ Banana 3 ◯ Banana # ◯ Banana ### ◯ Mango #####

4 **A bar chart whose axis starts at 68 shows a rise from 70% to 74% honestly.**

◯ True ◯ False

5 **Which should you check before believing a chart? Choose all that apply.**

☐ The title, axes and units ☐ Whether some months or groups were left out

☐ Where the bars start ☐ How bright the colours are

Answers · for the teacher

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1 **Where the axis starts**

An axis that doesn't start at zero makes small differences look big.

2 **Which fruit does our class like best? — Bar chart**
How did rainfall change this year? — Line chart
How is the school budget shared? — Pie chart
Do pupils who read more spell better? — Scatter chart

Groups, time, parts of a whole and two measurements: each kind of question has its chart.

3 **Banana ###**

On the second turn of the loop i is 1, so it prints "Banana" and "#" * 3, which is ###.

4 **False**

From 68, the bars are 2 and 6 units tall, so the new bar looks three times bigger.

5 **Where the bars start; The title, axes and units; Whether some months or groups were left out**

Cut axes, missing labels and missing data are the usual ways a chart misleads.

✎ **For you to think about**

Think of a chart you saw recently in a newspaper, on TV or on a poster. Was it honest? How could you tell?